MELINA BAGHER

Tehran-Iran   Last Update: August 14, 2026

Personal Website | LinkedIn page | Research Gate | Melina2bagher@gmail.com

+98-912-029-8521 | Google Scholar | GitHub | melina.bagher@sharif.edu

EDUCATION

MASTER OF SCIENCE IN BIOMEDICAL ENGINEERING — Sep. 2019 – Sep. 2022

Sharif University of Technology — *Tehran, Iran*

- **Thesis:** "Comparative Analysis of Haplotype Assembly Algorithms to Identify Optimal Methods"
- **Advisor:** Prof. Babak Hosseini Khalaj, Assoc. Prof. Mehran Jahed
- **GPA: 19.19/20 ➔ 4/4 (Ranked)**

BACHELOR OF SCIENCE IN BIOMEDICAL ENGINEERING — Sep. 2014 – Sep. 2018

Islamic Azad University, Science and Research Branch — *Tehran, Iran*

- **Thesis:** "Extraction of Fractal Dimension of Malignant and Benign Skin Lesions in Dermoscopic Images"
- **Advisor:** Asst. Prof. Saeid Rashidi
- **GPA: 19.10/20 ➔ 4/4 (Ranked)**

PUBLICATIONS

JOURNAL PAPERS

❍ **Bagher M.**, Karimzadeh R. , khalaj B.H., Jahed M; *A rapid heuristic algorithm to solve the single individual Haplotype assembly problem* ; AUT Journal of Electrical Engineering 55 (2), 191-206, 2023.

CONFERENCE PAPERS

❍ **Bagher M.**, Karimzadeh R. , khalaj B.H., Jahed M; *QuickHap: a Quick heuristic algorithm for the single individual Haplotype reconstruction problem* ; 29th National and 7th International Iranian conference on Biomedical Engineering 2022 (ICBME).

❍ Karimzadeh R., **Bagher M.**,Khodabakhsh A., Arabi H., Zaidi H; *Generative Adversarial-based Framework for Classification using Imbalance Data: Application to Pneumonia Detection in Chest Radiographs*; IEEE NSS and Medical Imaging Conference 2022.

SELECTED ACADEMIC PROJECTS

Artificial Intelligence and Deep Learning Projects/Tasks

GAN-Based Framework for Pneumonia Detection

GAN-based chest X-ray classification with feature-space augmentation. (*Published in IEEE NSS/MIC*)

Bone fracture classification using multi-level feature fusion

Multi-level feature fusion using InceptionV3-based transfer learning. (*Self-initiated project*))

CycleGAN for Image-to-Image Translation
implementation of CycleGAN for unpaired image-to-image translation.. (*Self-initiated project*)

Enhancing Adversarial Robustness in Convolutional Neural Networks
FGSM/PGD attacks and adversarial training in PyTorch. (*Course project*)

Image generation using VAE and Latent Diffusion pipleline
ext-to-image generation using VAE, U-Net, and latent diffusion (*Self-initiated project*)

License Plate Recognition using CNN-based Sequence Modeling
Developed a CNN-based model for Iranian license plate recognition. (*Self-initiated project*)

AI Agents and Automation Projects

AI-Powered Order Automation
Automated order summarization and email reporting using n8n, Google Sheets, Gmail, and LLMs.

Intelligent Calendar Assistant
Telegram-based AI Agent for Google Calendar management using n8n, LangChain, Whisper, and LLMs

AI Advertising Image Generation
Automated product image generation using n8n, OpenRouter, and multimodal AI models.

AI Data Analyst Agent
Automated data analysis and visualization from Google Sheets using n8n, LLMs, and QuickChart.

RAG-Based Knowledge Assistan
Built a RAG pipeline for document-based question answering using n8n, OpenAI Embeddings, and Vector Store

Classical Image Processing Projects/Tasks

Spinal Cord CT Registration Pipeline
CPD/ICP-based 3D CT registration using point clouds, evaluated with Dice, HD, and ASD (*Course project*)

Dermatoscopy Image Preprocessing and Segmentation
Skin lesion preprocessing and segmentation in MATLAB. (*B.Sc. thesis-related*)

Signal Processing Projects/Tasks

ECG Anomaly Detection using Autoencoder
TensorFlow-based anomaly detection on ECG5000 dataset. (*Deep Learning Course Project*)

Audio Signal Classification using Spectrogram-Based CNN
Speech command recognition using STFT spectrograms and CNN. (*Deep Learning Course Project*)

Computational Genomics and Bioinformatics Projects/Tasks

CNN for Single Individual Haplotype Reconstruction
PyTorch-based haplotype assembly with a custom loss function.(*M.Sc. thesis-related*)

NGS Data Simulation and Variant Calling Pipeline
NGS simulation and variant calling using BWA-MEM, GATK, Samtools, and BCFtools. (*M.Sc. thesis-related*)

Relevant COURSES

Selected Academic Courses

- **Medical Image Analysis and Processing** (Emad Fatemizadeh, Sharif University of Technology)
  Grade: 19.7/20
- **Biomedical Signal Processing** (Mohammad B. Shamsollahi, Sharif University of Technology)
  Grade: 18.9/20

Selected Online Courses

- **Introduction to ML and Deep Learning** (M H Rohban, Rayan.AI, Sharif Univercity of Technology)
- **Trustworthiness in DeepLearning** (M H Rohban, Rayan.AI, Sharif Univercity of Technology)
- **Supervised Machine Learning** (Andrew NG, DeepLearning.AI, Stanford University via Coursera)
- **Advanced Learning Algorithms** (Andrew NG, DeepLearning.AI, Stanford University via Coursera)
- **Deep Learning with TensorFlow and Keras** (Alireza Akhavanpour, Maktabkhooneh)
- **Practical AI Agent Development with n8n** (Soheil Tehrani Pour, Maktabkhooneh)

HONORS & AWARDS

- Ranked 1st among approximately 500 participants in the nationwide Ph.D. entrance exam for Biomedical Engineering (Bioelectric), 2023
- Ranked 2nd among approximately 900 participants in the nationwide Ph.D. entrance exam for Biomedical Engineering, 2026
- Ranked 39th among approximately 40,000 participants in the nationwide M.Sc. entrance exam for Electrical Engineering
- Member of National Elite Foundation
- Distinguished B.Sc. graduate

TEACHING EXPERIENCE

- **TEACHING ASSISTANT** | Medical Images Analysis and Processing | Supervisor: Assoc. Prof. Emad Fatemizadeh
  Department of Electrical Engineering, Sharif University of Technology *Fall 2021*
- **TEACHING ASSISTANT** | System Biology | Supervisor: Prof. Babak Hossein Khalal
  Department of Electrical Engineering, Sharif University of Technology *Spring 2020*
- **TEACHING ASSISTANT** | Electronics 2 | Supervisor: Assoc. Prof. Emad Fatemizadeh Keyvan Maghouli
  Department of Biomedical Engineering *Fall 2019*
- **TUTOR** | Engineering Mathematics, Differential Equations, Signals and Systems
  Mahan Institute of Higher Education *Summer 2019*

TECHNICAL SKILLS

- **Programming Languages:** Python, MATLAB, Bash
- **Python Libraries:** TensorFlow, Keras, PyTorch, Scikit-learn, Matplotlib, NumPy, SciPy
- **Version Control:** Git, GitHub
- **AI Agents and Automation:** n8n, AI Agent Workflows, familiar with LLMs, RAG, REST APIs, Webhooks
- **Hardware Simulator:** PSpice, Proteus
- **Typesetting:** TEX, Prezi, MS Office
- **Operating Systems:** Windows, Linux(Ubuntu)
- **NGS Tools:** GATK, BWA, Samtools, BCFtools, PBSIM, NanoSim, ART

LANGUAGES

- **English** TOEFL-iBT: 98 (R:28, L:21, S:23, W:26)
- **French** A1
- **Persian/Farsi** Native

RESEARCH INTERESTS

- Deep Learning, AI, Machine learning
- Image Processing, Computer Vision
- Signal Processing
- Computational Genomics, NGS data analysis

HOBBIES

SPORTS: Volleyball, Badminton, Table Tennis
MUSIC: Piano (Advanced), Guitar (Intermediate)

REFERENCES

Prof. Babak Hosseini Khalaj, *Department of Electrical Engineering, Sharif University of Technology*
Email: khalaj@sharif.edu

Asst. Prof. Saeed. Rashidi, *Department of Biomedical Engineering, Islamic Azad University, Science and Research Branch*
Email: rashidi@srbiau.ac.ir

Assoc. Prof. Emad Fatemizadeh, *Department of Electrical Engineering, Sharif University of Technology*
Email: Jahed@sharif.edu